

RUBY, AK, USA

WMO: 702627

Lat: 64.727N

Lon: 155.470W

Elev: 199

StdP: 98.96

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 26561

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.7	-35.1	-36.8	0.1	-33.4	-35.1	0.1	-31.6	13.1	-10.5	11.8	-12.6	1.1	80	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	25.7	15.6	23.5	14.4	21.6	13.8	16.6	23.4	15.5	21.4	14.6	20.0	2.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.1	18.1	12.9	9.5	17.2	12.2	9.0	16.6	47.1	23.4	43.8	21.4	41.3	20.0	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.4	8.7	7.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-38.5	28.9	3.8	1.7	-41.2	30.1	-43.4	31.1	-45.5	32.1	-48.3	33.3
			-34.8	17.8	1.5	1.3	-35.9	18.8	-36.8	19.5	-37.6	20.2	-38.7	21.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-19.9	-14.6	-11.3	-0.9	8.7	14.5	15.5	12.3	6.6	-2.9	-13.9	-17.2
	DBStd	14.07	10.18	7.57	6.98	6.02	5.44	3.47	3.53	3.09	4.16	5.17	6.64	7.88
	HDD10.0	4782	927	687	661	328	87	3	2	12	115	399	718	844
	HDD18.3	7394	1185	921	919	578	300	122	102	188	353	657	968	1102
	CDD10.0	454	0	0	0	0	47	139	173	83	12	0	0	0
	CDD18.3	26	0	0	0	0	1	9	15	1	0	0	0	0
	CDH23.3	196	0	0	0	0	14	63	114	4	0	0	0	0
	CDH26.7	31	0	0	0	0	1	12	18	0	0	0	0	0
Wind	WSAvg	2.7	3.0	4.0	3.2	3.2	2.3	2.1	2.0	2.2	2.7	2.7	2.8	2.9
Precipitation	PrecAvg	487	31	28	21	17	23	44	64	77	59	45	40	37
	PrecMax	604	84	84	70	46	44	74	113	119	127	138	129	94
	PrecMin	333	4	1	0	2	2	11	21	35	20	8	3	6
	PrecStd	78	20	18	15	12	10	17	22	22	31	25	29	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	0.6	4.7	14.3	25.5	28.4	28.9	23.7	18.6	9.0	1.6	-0.3
		MCWB	0.7	-1.3	1.1	6.8	13.2	16.3	17.0	15.8	11.9	4.8	-0.5	-1.4
	2%	DB	-1.3	-0.8	2.5	12.0	22.5	25.0	26.4	21.3	16.2	6.3	-1.0	-2.4
		MCWB	-2.1	-2.1	-0.8	5.2	12.1	14.8	16.1	14.6	10.5	4.5	-1.7	-3.1
	5%	DB	-3.5	-2.5	0.5	9.9	19.3	23.1	24.2	19.1	14.4	4.9	-3.3	-4.7
		MCWB	-4.2	-4.0	-2.1	4.1	10.6	13.8	15.7	13.6	9.9	3.3	-4.1	-5.5
	10%	DB	-6.6	-4.5	-1.6	7.6	16.9	21.0	22.0	17.6	12.6	3.4	-5.5	-6.7
		MCWB	-7.3	-5.6	-3.7	2.9	9.3	13.0	14.5	12.9	9.1	2.1	-6.2	-7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	-0.7	2.2	7.2	13.8	17.8	18.4	17.0	12.3	6.2	0.1	-1.0
		MCDB	1.9	0.0	4.6	13.8	23.8	25.7	26.6	22.2	17.2	7.2	0.8	-0.2
	2%	WB	-2.0	-2.0	-0.4	5.6	12.4	15.9	17.1	15.4	11.3	4.8	-1.6	-3.0
		MCDB	-1.3	-0.8	2.0	11.1	21.0	22.8	24.7	19.3	15.0	6.1	-0.8	-2.2
	5%	WB	-3.9	-3.8	-2.1	4.4	11.2	14.7	15.9	14.5	10.3	3.4	-4.0	-5.3
		MCDB	-3.2	-2.6	0.3	9.3	18.4	21.3	22.4	18.1	13.4	4.6	-3.2	-4.7
	10%	WB	-6.7	-5.6	-3.8	3.1	10.0	13.7	15.1	13.6	9.6	2.3	-6.1	-7.2
		MCDB	-6.1	-4.6	-1.5	7.3	15.7	19.8	21.0	16.8	12.5	3.3	-5.5	-6.6
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.2	8.7	9.0	9.5	9.1	8.3	7.3	6.5	4.3	4.9	4.9
		MCDBR	5.6	6.3	9.0	10.5	12.2	11.2	11.2	9.7	9.3	5.2	5.6	4.9
		MCWBR	5.1	5.5	6.6	5.9	5.6	4.9	4.8	4.6	5.1	3.8	5.3	4.5
	5% WB	MCDBR	5.8	6.6	8.5	9.9	11.3	10.0	10.1	8.0	7.8	4.6	5.5	4.9
		MCWBR	5.2	5.9	6.4	5.7	5.2	4.8	4.7	4.3	4.6	3.6	5.3	4.6
Clear-Sky Solar Irradiance	taub		0.170	0.245	0.250	0.314	0.332	0.365	0.367	0.365	0.303	0.264	N/A	0.174
	taud		1.939	2.205	2.302	2.216	2.434	2.358	2.397	2.409	2.559	2.461	N/A	1.930
	Ebn at Noon		393	677	856	855	878	848	835	799	796	670	N/A	125
	Edh at Noon		23	54	76	108	96	107	100	90	62	43	N/A	8
All-Sky Solar Radiation	RadAvg		0.17	0.84	2.64	4.61	5.30	5.48	4.68	3.51	2.23	0.91	0.25	0.05
	RadStd		0.02	0.08	0.20	0.32	0.53	0.41	0.36	0.33	0.29	0.12	0.04	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+1.08	N/A	+0.85	N/A	N/A	N/A	-319	-384	+63	N/A

Nomenclature: See separate page